

Does geographic remoteness increase local financial dependence? Evidence from Mongolia

Scientific method framework

The research investigates isolated provincial governments that rely on central government grants besides revenue and expenses. Mongolia historically maintained a central fiscal system; the budget law changed in 2013, increasing the opportunity to localities to generate their own revenues based on their resources. Considering Mongolia's specific demographic and geographical characteristics, the country has a total population of 3.4 million, a territory of 1.5 million square km, and 21 provinces as second tier of administrative level. These provinces are largely engaged in seasonal agriculture and year-round livestock herding activities, which are core to the country's economic and sustainable development. Over ninety percent of the country's GDP comes from the mining sector, while localities depend on livestock and a small portion of the agricultural sector.

Local financial dependence and autonomy depend on state fiscal support (grants) and are measured by the ratio of tax and non-tax revenue to expenses, depending on provincial population, size, development index and tax initiatives. The government is sometimes reported to provide additional grants to remote provinces based on distance from the economic hub and the effects of the geographical characteristics and the weather hazards such as dzud (severe winter disaster), droughts that increase livestock mortality and reduce local revenues. However, remoteness and distance have not been formally incorporated into the budget law as explicit criteria for grant allocation supported by a clear theoretical and methodological framework.

Theoretical paths concerning public goods, equal distribution, stability started in the 1950s with Samuelson's (1954) theory of public expenditures rather than taxation, and were expanded by

Tiebout's (1956) theory of local expenditures, mobility. Musgrave (1959) and Oates (1972) discussed allocation, distribution and stabilization of functions, and the role of local government and intergovernmental grants. Local financial indicators define the degree of local authorities' dependence on, autonomy and independence from state or central governments.

After the unification of the European Union, the majority of former socialist countries shifted from centralized to decentralization of budget planning, and considered more variables from different disciplines to define an optimal ratio of local economic capacity Beer-Toth (2009). Borck & Owings (2003) considered spatial spillover effects on intergovernmental grants.

Problem definition

Human capital (education hub) and focal economic activity are concentrated in the capital city, attracting a substantial share of the country's population. Mongolia also has limited economic diversification and occupies a landlocked position between Russia and China. Based on provincial coordinates and author's calculation from previous assignments, interprovincial distances range from approximately 300-1250 km, with around 75 percentiles of measured distances falling within 1000 km. These distances reflect the country's vast territory and low population density. Some provinces also remain geographically isolated from the capital city far beyond the range. Against this background, how are the distances between provinces and from the capital city associated with local financial capacity and the allocation of central government grants?

Research questions and hypothesis

H1: Local financial dependence reflects heavy reliance on central government grants and remoteness may be one of the factors (positive expected results)

H2: Isolated provinces are more likely to receive greater state fiscal support than provinces closer to the capital city (positive expected results)

Data sources identification/ accessibility

The dataset comprises panel data covering 21 provinces over 6 years, including local revenue, local expenditures and central government grants obtained from the National Statistical Office of Mongolia (NSO). These data are combined with a geospatial shapefile (Geodata) to identify provincial coordinates and calculate Euclidean distance matrices between provinces, supporting spatial analysis and provincial clustering. A complete dataset would contain 126 province-year observations. Additional local economic indicators, including local population, local GDP per capita, local development index and size are also available for potential inclusion. Ulaanbaatar is excluded from the provincial sample because of its distinct role as the country's fiscal and administrative hub, but it remains the reference location for measuring distance to the capital. The statistical data are obtained from an accredited source of the National Statistics Office of Mongolia, and the website and the URL are publicly accessible.

Analytical methods and Validation approach

The following theoretical equation will be applied to calculate local financial ratios.

$$\begin{aligned} \text{local financial capacity} &= \frac{\text{local revenue}}{\text{local expenditure}} \\ \text{local financial dependence} &= \frac{\text{grants}}{\text{local revenue} + \text{grants}} \end{aligned}$$

In the regression model, the distance coefficient measures the difference in the financial ratio associated with each additional 100 km from Ulaanbaatar. The proposed approach is pooled

OLS. Although the final model specification remains under consideration. The preliminary specification is as follows:

$$LFD_{it} = \alpha + \lambda_t + \beta \left(\frac{D_i}{100} \right) + \varepsilon_{it}$$

where LFD_{it} denotes the local financial dependence of province i in the year t , α is the intercept, λ_t represents year effects, β measures the difference in the financial dependence ratio associated with each additional 100 km from the city, D_i is the distance in kilometers from province i 's to the city and ε_{it} is the error term.

Related studies have used first-stage F statistics to assess strength, Anderson-Rubin tests for the inference robust to weak instruments, and placebo regressions to examine alternative explanations. For this study, model fit will be assessed using R^2 . The Breusch-pagan panel LM (Lagrange Multipliers) test will assess whether a province level random effect component is present. The Jargque-Bera test may also be reported as checking residual normality.

Value proposition and success metrics

If the study finds that remoteness is associated with local financial dependence, policy makers could consider incorporating distance-based criteria into the budget law. The policy implication could extend beyond grant allocation to exploring the establishment of additional regional municipalities. Further research could also examine average temperatures in the western and northern parts of the country, along with other geographical characteristics.

Technical implementation and milestones

Python will be used with pandas and numpy for data preparation, statsmodels for pooled OLS and Matplotlib and seaborn for spatial and geographical visualization.

Week 1- develop the theoretical sections and merge observations and clean the available dataset

Week 2, 3- conduct analysis and prepare reports

Week 4- perform validation and robustness checks

Week 5,6- complete the final research paper and reproducible research package

References

Beer-Tóth, K. (2009). Local financial autonomy in theory and practice: The impact of fiscal decentralization in Hungary [Doctoral dissertation, University of Fribourg]. *FOLIA-Fribourg Open Library and Archive*

<https://folia.unifr.ch/unifr/documents/301442>

Borck, R., & Owings, S. (2003). The political economy of intergovernmental grants. *Regional Science and Urban Economics*, 33(2), 139–156.

[https://doi.org/10.1016/S0166-0462\(02\)00005-4](https://doi.org/10.1016/S0166-0462(02)00005-4)

Musgrave, R. A. (1959). *The theory of public finance: A study in public economy*. McGraw-Hill. Google Books.

Oates, W. E. (1972). *Fiscal federalism*. Harcourt Brace Jovanovich. Google Books

Samuelson, P. A. (1954). The pure theory of public expenditure. *The Review of Economics and Statistics*, 36(4), 387–389. <https://doi.org/10.2307/1925895> bibbase.org

Tiebout, C. M. (1956). A pure theory of local expenditures. *Journal of Political Economy*, 64(5), 416–424. <https://doi.org/10.1086/257839>